

2 Literature Review

Climate stress testing draws on two literatures that grew up apart. One produces scenarios reaching out decades; the other produces valuations re-marked monthly. What follows traces the attempts to join them, moving from the supervisory practice that created the demand, through the climate-economy models that supply the scenarios and the competing frameworks that translate them into economic quantities, to the input–output tradition, the long struggle to extend it across borders, and the empirical relations that turn a real shock into a market price.

The terms were set by Carney (2015), whose address at Lloyd's named the obstacle a *tragedy of the horizon*: the costs of climate change fall beyond the planning horizons of the business cycle, the political cycle and a central bank's mandate, so the incentive to act expires before the damage becomes visible. His remedy — forward-looking scenario analysis in place of realised-loss accounting — is what the decade of supervisory work that followed attempted to carry out. The largest of those exercises, the European Central Bank's economy-wide climate stress test (ECB, 2022), established the shape the field has kept: a scenario taken from outside, mapped onto counterparty exposures, propagated, and revalued.

Whether that shape works is assessed by Acharya et al. (2023), and their verdict is mixed. They find transition risk treated as an exogenous shock when it is in substance a dynamic policy choice, no feedback between climate and economy, and few compound scenarios in which several risks arrive at once. Their fourth objection cuts deeper than the other three. The exercises are not reproducible by the institutions being tested, because the models are large, proprietary and opaque; a bank required to hold capital against a supervisory number cannot rebuild that number, and so cannot discover which of its own exposures the number is about.

The academic literature that grew alongside those exercises begins with the network stress test of Battiston et al. (2017), whose contribution is as much procedural as empirical. Sorting economic activities into climate-policy-relevant sectors, mapping bank and investor portfolios onto them, and pushing a policy shock through the network of interbank exposures, they find that exposure arriving indirectly through counterparties is comparable in size to direct exposure, and that the affected share of bank loan books is of the same order as bank capital. The general lesson is that indirect exposure cannot be inferred from direct exposure and has to be computed through an explicit network.

Two lines run on from there. Whether markets have already priced the risk is asked by Bolton and Kacperczyk (2023), who estimate a carbon premium across some 14,000 firms in 77 countries and find higher returns for firms with higher emissions, the premium larger where domestic climate policy is stricter — which is evidence that transition risk sits partly in prices already, and a limit on what any stress test can claim to be uncovering. A different move is made by Desnos et al. (2023), who convert a deterministic stress test into a stochastic climate value-at-risk on the argument that a scenario is a draw from a distribution rather than a state of the world.

Both lines take the scenario as given, and the scenarios come from integrated assessment models. The lineage runs through DICE, which couples a Ramsey growth model to a simple carbon cycle and a quadratic damage function mapping temperature to lost output, recalibrated most recently by Barrage and Nordhaus (2024). Of the components in that structure the damage function rests on the least evidence, and the criticism of it is well developed. The sharpest statement is Pindyck's (2013): the functional forms have, on his reading, no theoretical or empirical basis for their curvature at high temperatures, so their outputs should be treated as illustrations rather than as estimates. A separate objection is raised by Dietz and Stern (2015), who let damage erode the capital stock and the growth rate rather than only the level of output and find the implied social cost of carbon rising sharply, supporting deeper cuts than the standard calibration recommends. Set against those arguments, the coefficient of Swiss Re Institute (2024) — roughly five times the DICE central value at any temperature — reads less as a rival estimate than as the other end of an unsettled range.

Splitting an aggregate loss across economies calls for a different instrument, and the one in common use is the vulnerability index of the Notre Dame Global Adaptation Initiative (2024), which folds exposure, sensitivity and adaptive capacity into a single cross-economy score. It ranks rather than measures: it can order economies by vulnerability, but says nothing about how much any of them loses. That is why work using it pairs it with a damage function instead of substituting it for one.

The scenario set supervisors have converged on is published by the Network for Greening the Financial System (2024), and its architecture decides what users must supply for themselves. Transition pathways come from three integrated assessment models — GCAM, MESSAGEix-GLOBIOM and REMIND-MAGPIE — whose carbon-price and energy trajectories are handed to the macroeconomic model NiGEM for macroeconomic variables, with chronic physical risk entering through an aggregate damage function and acute risk through hazard-specific models. Because the macroeconomic layer is attached to the integrated assessment layer rather than derived from it, the published scenarios carry no consistent mapping from a carbon price to a sectoral cost. Filling that gap is what every sectoral application of these scenarios must do, and it has been done in four broadly different ways.

A classification of the four is offered by Hafner et al. (2020). Environmental computable general equilibrium models impose optimising behaviour and market clearing, and are the conventional instrument for carbon-tax incidence; twelve of them are compared on common unilateral-policy and border-adjustment experiments by Böhringer, Balistreri and Rutherford (2012), in a study that says as much about the family as about the policy. Leakage rates cluster between five and twenty per cent, but the spread across models is driven by the substitution elasticities each one assumes — elasticities estimated with wide uncertainty, and invisible in an input–output table.

Macroeconometric models keep an input–output core and replace optimisation with estimated behavioural equations, which lets the economy carry persistent slack. That capacity is used by Mercure et al. (2018) to value stranded fossil-fuel assets at a discounted global wealth loss of one to four trillion dollars, a figure that depends on the economy failing to clear and could not have come out of a general-equilibrium treatment of the same shock.

Interlocking balance sheets are tracked instead by ecological stock-flow consistent models, which makes financial feedback explicit rather than assuming it away. Working in such a model, Dafermos, Nikolaidi and Galanis (2018) show climate damage raising corporate defaults and deflating asset prices, with the resulting instability feeding back to worsen the damage itself. Nothing discussed above contains that loop.

Building upward from heterogeneous firms and households, ecological agent-based models arrive at different magnitudes altogether. Damages well above those of standard integrated assessment models are reported by Lamperti et al. (2018) under comparable scenarios, with growth tipping from a self-sustaining path into stagnation and volatility — a direct challenge to the calibrations above, and a suggestion that the smoothness of the damage function matters more than its coefficient.

The pattern across the four is consistent. Substitution, disequilibrium, financial feedback and heterogeneity each turn out to matter; none survives in a fixed-coefficient input–output system; and three of the four literatures point the resulting bias in the same direction, toward understatement.

The input–output core that the macroeconometric family retains is older than any of these, and it acquired an environmental branch early. An economy was first described as a system of linear production relations by Leontief (1936), each sector's requirement a fixed combination of every other sector's output, so that with $\mathbf{A}$ the matrix of technical coefficients and $\mathbf{f}$ final demand, equilibrium output solves $\mathbf{x} = \mathbf{A}\mathbf{x} + \mathbf{f}$ and hence $\mathbf{x} = (\mathbf{I} - \mathbf{A})^{-1}\mathbf{f}$. The condition under which that inverse exists and means anything was established by Hawkins and Simon (1949) in terms of the leading principal minors of $\mathbf{I} - \mathbf{A}$, which turns what looks like a computational convenience into a statement about whether an economy is productive.

Pollutants were added as rows of the same accounting system by Leontief (1970), so that emissions are attributed to the production generating them and travel with the goods themselves; that paper begins the environmentally extended input–output literature and is the ancestor of the sectoral carbon intensities in use today. The modern synthesis is Miller and Blair's (2022), and three of their treatments recur throughout applied work: the price dual, which transposes the same matrix to ask how far prices move when costs change rather than how much output a demand shock requires; the inter-country generalisation, in which the flow matrix is blocked by region and the off-diagonal blocks hold trade; and the aggregation problem, since merging units is demonstrably not neutral and the direction of the bias depends on how unlike the merged units are.

Extending that apparatus across borders proved harder than extending it across sectors, and for fifty years the multi-regional literature circled one obstacle: the inter-regional flows are the object one needs and the object one rarely has. The exact formulation was written down by Isard (1951), in which a_{ij}^r records how much of good i from region r sector j in region s requires. It is also the least usable, since it demands a full bilateral flow matrix that statistical agencies do not collect.

Two responses went opposite ways around the missing data. Exact accounting was preserved at the cost of behavioural detail by Chenery (1953) and Moses (1955), who assume that every user of a good within a region draws on the same mix of origins, so that one trade coefficient serves all using sectors and $a_{ij}^r = t_i^r a_{ij}^s$; that needs only regional trade totals, at the cost of assuming a car plant and a hospital in the same region buy their steel from the same places. The missing flows were instead estimated by Leontief and Strout (1963), who treat regional output as entering a national pool and suppose that bilateral flows obey a gravity relation,

$$z_i^r = \frac{x_i^r d_i^s}{x_i^{\bullet} Q_i^r},$$

with Q_i^r a friction term falling in transport cost. Since the frictions have to reproduce known row and column totals, the system is solved iteratively, by the biproportional fitting brought into input–output work by Stone (1961). The cost of that elegance is that the trade pattern becomes a model output, so any result turning on how a shock crosses a border inherits the gravity specification's error on top of the model's own.

The obstacle dissolved once the flows were measured instead of inferred. The programme that assembled global multi-regional tables from national accounts and bilateral trade statistics through the 2000s is reviewed by Tukker and Dietzenbacher (2013), and it produced four major databases within a few years of each other: WIOD, documented by Timmer et al. (2015), alongside EXIOBASE, Eora and the GTAP-based compilations. Whether findings depend on which database is chosen was tested by Moran and Wood (2014), who harmonise satellite accounts and compare consumption-based carbon accounts across them, finding agreement within roughly ten per cent for most major economies and attributing what remains to industry aggregation, emissions data and modelling assumptions. Belonging to the same generation, the inter-country tables of the OECD (2025) supply the off-diagonal blocks directly.

Beyond propagation, measured inter-country tables make consumption-based accounting possible: attributing emissions to the final demand that occasions them rather than to the territory where they are released. How wide the gap between the two accounts runs was measured by Davis and Caldeira (2010), who find that much of what developed economies consume was produced elsewhere. For anyone ranking economies by climate relevance, the implication is that a ranking on trade weight and a ranking on embodied emissions are not the same list.

What such a system is usually asked to price is a carbon charge, whose design has a history of its own, beginning with the comparison of prices and quantities under uncertainty in Weitzman (1974). That debate need not be settled here, but a caution from Green (2021) is worth carrying: reviewing ex post evaluations of carbon pricing, he concludes that measured emissions reductions have been modest against expectations. A carbon price is a contested policy variable rather than a fact of nature.

Given a price, who bears it has mostly been worked out through the input–output price model. One is applied to a carbon tax driven by the NGFS scenarios by Kay and Jolley (2023), who find price increases of ten to thirty per cent in carbon-intensive industries under a two-hundred-dollar tax, for a single economy. The cost-push formulation is developed further by Roncalli and Semet (2024), who follow a tax from the sectors it is levied on to the sectors that end up paying it, and separate the charge on a firm's own emissions from the charge embodied in what it buys. Two of their findings matter for anything built on that apparatus: the indirect burden is often the larger of the two, and the split between what a firm absorbs and what it passes on is genuinely open, turning on market structure, contract length and demand elasticity, none of which an input–output table records.

That openness is confirmed rather than closed by the pass-through literature. Theory and measurement are surveyed by RBB Economics (2014), who find rates varying widely with market structure and with the kind of cost shock involved. A stronger claim is made by Weber and Wasner (2023), who argue that under general cost pressure firms with market power can pass on more than their cost increase, which makes rates above one a live possibility rather than an arithmetic ceiling.

Whether a unilateral price merely moves emissions elsewhere is the leakage question, and the literature answers it by comparing models rather than trusting any one, because the answer turns on trade and energy-market elasticities that no single framework pins down. Already met above for what it shows about general-equilibrium models, the study of Böhringer, Balistreri and Rutherford (2012) is the reference point on substance too: leakage clusters between five and twenty per cent, and border carbon adjustment reduces it without removing it. In that literature's terms, a border adjustment is a tariff differentiated by embodied carbon.

An industrial cost becomes a financial quantity only through a further chain of relations, each borrowed from a separate empirical literature, and the first link is the shakiest. New-Keynesian Phillips curves are estimated across OECD economies over 1995–2020 by Moessner (2022), who finds that a ten-dollar rise in the carbon price raises headline inflation by roughly 0.08 percentage points. That estimate is disputed. Working with carbon taxes actually implemented in Europe and Canada over three decades, Konradt and Weder di Mauro (2023) find dynamic effects on headline inflation indistinguishable from zero, and read their evidence as relative price change — energy dearer, other things cheaper — rather than inflation. The two are only partly reconciled by Bauer and Känzig (2024), who find carbon pricing moving inflation *expectations* even where realised inflation barely moves. Adopting a point estimate here means taking a side in a live dispute.

The rest of the chain is firmer. The monetary reaction function is Taylor's (2007). The term structure follows Hull and White (1994), whose one-factor model gives the shift at maturity $\Delta\tau$ as a deterministic multiple $B(\tau)/\tau$ of the short-rate shift, independent of volatility, so that a curve displacement follows without calibrating a volatility surface. On the two exchange-rate parities the standard reference remains Sarno and Taylor (2002), candid that relative purchasing power parity does badly at short horizons while covered interest parity is close to an arbitrage identity.

Those relations are assembled into what its authors call an ensemble by Berrahoui, Kenyon, Macrina and Nathanael (2025): a short sequence of individually simple, individually auditable steps running from a climate scenario to a set of market shocks, two of them stated as propositions and proved — the allocation of aggregate damage across sectors, and the mapping of a short-rate shift onto the term structure. The design descends from earlier work by the same authors on pricing carbon consistently inside financial instruments (Kenyon, Macrina and Berrahoui, 2022). Its distinguishing property is the one Acharya et al. (2023) found missing from supervisory practice: every relation is visible, so a disagreement about the answer can be traced to a disagreement about one relation. Its cost is realism, in precisely the dimensions the four alternative model families were built to capture.

It is also, as published, a single-economy framework calibrated on the United Kingdom, and that restriction has two consequences given everything above. Emissions, and the cost of pricing them, demonstrably cross borders through production chains — that is the finding of Leontief (1970) and of the consumption-based accounting tradition after Davis and Caldeira (2010) — and a one-country model has nowhere for that cost to go. The exchange rate, the cross-currency basis and every other relative price, meanwhile, are not merely harder to compute in such a model; they are undefined in it.

The gap is therefore narrow and specific. An auditable stress-testing chain with proved internal relations exists for one economy (Berrahoui et al., 2025). A mature multi-regional input–output apparatus exists (Miller and Blair, 2022; Tukker and Dietzenbacher, 2013), together with measured inter-country tables that remove the estimation step the Leontief–Strout tradition had to live with. Input–output treatments of carbon-tax incidence exist for single economies (Kay and Jolley, 2023; Roncalli and Semet, 2024). Nothing joins them, and until something does, the quantities that exist only between economies stay outside the reach of an auditable stress test.

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